

Title : Effect of Pose-Diverse Training on Cross-Pose Face Recognition and Learned CNN Representations

Selected Track: Track A — Architecture and Representation

Research Question:

Does pose-diverse training improve cross-pose CNN face recognition and reduce pose-induced representation shift compared with frontal-only training?

Hypothesis:

H1: Frontal-only training will show decreasing recognition performance as yaw increases, while pose-diverse training will improve non-frontal recognition.

H2: Same-identity embedding similarity to the frontal representation will decrease with yaw, but the pose-diverse model will preserve higher similarity.

Dataset and Data Split:

A filtered Multi-PIE subset with **153 identities**, yaw angles from 0° to $\pm 90^\circ$, and 20 illumination conditions per pose will be used. Eyeglass images are excluded. The split is performed by illumination condition and kept fixed across experiments:

- Train: 12 illumination conditions
- Validation: 4 illumination conditions
- Test: 4 illumination conditions

Baseline:

Model A: CNN trained only on frontal (0°) face images, using **12 images per identity (1,836 total)**.

Proposed Comparison:

Model B: Same CNN architecture and training-image budget as Model A, but trained using $0^\circ, \pm 30^\circ, \pm 60^\circ$ yaw images. The training allocation per identity is 4 frontal images and 2 images from each non-frontal pose.

Metrics:

Top-1 accuracy, Macro-F1, accuracy by absolute yaw angle, training/validation learning curves, and same-identity frontal-to-non-frontal embedding cosine similarity. Model B will also be compared across seen, unseen-interpolation, and extreme-extrapolation poses.

Expected Risks:

Extreme yaw may cause severe self-occlusion and reduce available identity information. The model may also perform well on observed poses but generalize poorly to unseen viewpoints. Results from a controlled, closed-set dataset may not directly represent real-world performance.

Model	Training Poses	Images / ID	Evaluation
A: Baseline	0°	12	Common held-out test set
B: Pose-diverse	$0^\circ, \pm 30^\circ, \pm 60^\circ$	12	Common held-out test set

Controlled variables: CNN architecture, preprocessing, optimizer, training budget, validation/test split, and training protocol.